



PRODUCT & OPERATING REQUIREMENTS

Digital Asset Management

Business Requirements Document

DAM Portal | Superadmin Portal

VERSION 1.0

DATE 21 August 2026

PREPARED FROM Application codebase and project specifications

CLASSIFICATION Internal - Confidential

A unified business specification for organizational asset management and platform operations

Contents

- 01 Executive Summary**
- 02 Scope, Stakeholders & Business Context**
- 03 Business Requirements**
- 04 Non-Functional Requirements**
- 05 Current State Analysis**
- 06 Future-State Operating Design**
- 07 Workflow Diagrams & User Journeys**
- 08 Data Flow & Data Requirements**
- 09 Technology & Architecture**
- 10 Project Plan, Governance & Quality**
- 11 Benefits Realisation**
- 12 Appendices**

Navigation note: Word's Navigation Pane can be used to jump between all numbered sections and requirements.

01 - Executive Summary

Document Overview

This Business Requirements Document (BRD) defines the business objectives, product scope, functional requirements, non-functional requirements, operating workflows, data requirements, architecture, governance, and acceptance framework for the Digital Asset Management (DAM) platform.

The platform consists of two connected product surfaces:

1. **DAM Portal** - the organization-facing workspace used by administrators, Department Heads, Team Leads, members, and interns to onboard, organize, upload, govern, find, preview, process, share, publish, and manage digital assets.
2. **Superadmin Portal** - the platform-operations workspace used to approve organizations and Google Drive connections, administer organizations and users, control organization feature access, oversee nomenclature policies, and monitor platform-level totals.

This BRD is derived from the project codebase and its supporting specifications. It expresses the complete product requirement as a unified target specification. Commercial values or policy decisions that are not established in the project are marked **[Stakeholder input required]**.

Business Problem Statement

Organizations commonly store creative and operational files across local drives, personal cloud accounts, shared folders, and disconnected department repositories. This creates recurring business problems:

1. **Fragmented asset storage** - teams cannot reliably determine where the authoritative file is stored.
2. **Inconsistent organization** - folders, filenames, and metadata differ by team and user.
3. **Weak governance** - file operations and folder creation are difficult to control consistently across departments.
4. **Slow retrieval** - users spend unnecessary time browsing folder structures or searching by incomplete filenames.
5. **Duplicate assets** - repeated uploads increase storage use and create uncertainty about the correct asset.
6. **Limited operational control** - platform operators need a separate, secure surface for organization approvals, user administration, and tenant configuration.
7. **Storage constraints** - organizations require flexibility to use platform storage, Google Drive, or their own supported cloud object storage.
8. **AI-readiness gaps** - documents and media need structured processing so approved users can use Retrieval-Augmented Generation (RAG) and semantic retrieval workflows.

Business Opportunity

The DAM platform provides a governed, organization-aware asset layer that combines familiar file management with department hierarchy, controlled nomenclature, configurable permissions, multiple storage backends, document preview, metadata, and RAG preparation. The business opportunity is to

reduce asset-handling effort while improving consistency, discoverability, access control, and operational accountability.

Product Vision

Vision statement: *Enable every organization to maintain a secure, governed, searchable, storage-flexible library of digital assets while giving platform operators clear control over tenant onboarding and service configuration.*

Strategic Goals

ID	Goal	Business Measure	Target
G-01	Establish one governed asset library per organization	Assets managed through the shared organization workspace	100% of in-scope organizational assets
G-02	Standardize asset ingestion	Uploads evaluated against configured naming and file-format rules	100% when governance is enabled
G-03	Reduce time to locate assets	Median time from search to asset open/download	[Stakeholder input required]
G-04	Enforce least-privilege access	File actions resolved through role, department, and user-level controls	100% of protected operations
G-05	Support storage choice	Supported organization storage modes	Platform/local, Google Drive, AWS S3, Cloudflare R2, and Google Cloud Storage
G-06	Prepare supported content for knowledge retrieval	Eligible assets processed into RAG artifacts	[Stakeholder input required]
G-07	Centralize platform operations	Organization onboarding and lifecycle actions handled in Superadmin	100% of organization requests

Key Performance Indicators

- Median asset search-to-open time.
- Upload success rate and nomenclature compliance rate.
- Duplicate upload detection rate.
- Preview success rate by supported file type.
- RAG processing completion and failure rates.
- Folder-request approval turnaround time.

- Active organization and approved-user counts.
- Storage consumed per organization.
- Google Drive connection approval turnaround time.
- Percentage of privileged actions authorized through configured permission rules.

Critical Success Factors

1. Strong tenant isolation at every application and data-access boundary.
2. Consistent permission enforcement in both the user interface and server APIs.
3. Reliable storage routing across platform, Google Drive, and BYOS configurations.
4. Clear, usable governance controls for administrators and Department Heads.
5. Predictable upload behavior for individual files, batches, and directory trees.
6. Stable preview and RAG-processing flows for supported content types.
7. Secure separation between the DAM Portal and Superadmin Portal.

02 - Scope, Stakeholders & Business Context

In Scope

DAM Portal

- Authentication and user onboarding through Google or GitHub OAuth.
- Organization creation or organization-join request flows.
- Organization approval and pending/suspended access states.
- Shared organization asset library and folder hierarchy.
- Grid, compact-grid, list, and table library views.
- File, multi-file, and directory-tree upload.
- Explicit upload destination selection.
- Department-aware upload routing.
- File-format restrictions and file/folder nomenclature rules.
- Duplicate detection and deterministic collision handling.
- Folder creation and folder-request approval.
- File and folder open, preview, download, copy, move, rename, delete, restore, share, favorite, publish, and metadata actions.
- Recent, Favorites, Shared, Published, and Trash collections.
- Keyword, filter, faceted, smart, and RAG-supported search.
- Asset metadata, tags, and custom taxonomy fields.
- Organization, department, user, role, and permission administration.
- Template folder structures.
- Google Drive selection, approval, listing, upload, download, rename, delete, and link import.
- BYOS connection verification and organization storage configuration.
- RAG processing for supported files and progress reporting.
- Responsive layout and dark/light display modes.

Superadmin Portal

- Separate Superadmin authentication and session management.
- Platform dashboard with organization, user, storage, and Google Drive totals.
- Organization-request review and approval or rejection.
- Google Drive hosting-request review and approval or rejection.
- Organization listing, detail, suspension, reactivation, and deletion.
- Organization feature-access controls.
- Cross-organization user listing, editing, suspension, and deletion.
- Organization nomenclature oversight by department.

Out of Scope

- Asset version history.
- AI auto-tagging and optical character recognition.
- Multi-stage asset approval workflows.

- External brand portals or public collections beyond the existing publish/share capability.
- Digital rights management and licence-expiry enforcement.
- Media transformation services such as cropping, transcoding, rendition generation, or CDN optimization.
- Product analytics dashboards beyond the platform totals defined for Superadmin.
- Notification and messaging systems.
- Billing, subscription plans, invoicing, or payment processing.

Stakeholders

Stakeholder	Primary Interest	Key Responsibilities
Platform Owner	Service governance and product direction	Approves policy, funding, scope, and release decisions
Superadmin	Cross-tenant platform operations	Reviews requests, administers tenants and users, controls feature access
Organization Admin	Organization setup and governance	Configures departments, roles, storage, nomenclature, taxonomy, and users
Department Head	Department-level governance	Manages department permissions, members, folder requests, and nomenclature where authorized
Team Lead	Day-to-day team coordination	Uploads and organizes assets; requests governed folders; may receive elevated permissions
Team Member	Asset contribution and use	Uploads, searches, previews, downloads, shares, and manages assets as permitted
Intern	Restricted asset use	Performs explicitly permitted asset operations
Infrastructure / Operations	Availability and data protection	Deploys, monitors, backs up, restores, and maintains the runtime
Security / Compliance	Access and data safeguards	Reviews identity, authorization, credential handling, and retention controls

Assumptions and Dependencies

- Each user has at most one active organization context in a session.
- Each organization has one shared logical DAM bucket named `org`.
- Organization users are associated with an organization and may be associated with a department.
- OAuth provider availability is required for standard user authentication.
- Google Drive features depend on valid Google OAuth tokens and granted Drive permissions.
- BYOS features depend on valid organization-supplied cloud credentials and an accessible bucket.
- RAG processing depends on configured parsing, embedding, and vector-storage services.
- Platform deployment may use Cloudflare-compatible services or a Node/GCP runtime with persistent storage.

03 - Business Requirements

Requirement Conventions

- **Must Have** - required for the product to meet its core business purpose.
- **Should Have** - materially improves governance or usability and is expected within the product scope.
- **Could Have** - useful capability that may be scheduled after mandatory requirements.
- Acceptance criteria are written as observable business outcomes and do not prescribe a single implementation.

A. DAM Portal - Identity, Onboarding & Workspace

FR-100: OAuth Authentication

Field	Detail
Requirement ID	FR-100
Description	Users shall sign in to the DAM Portal through supported Google or GitHub OAuth providers.
Business Rules	A successful provider callback shall resolve or create the user identity and establish a protected application session. Unauthenticated users shall be redirected to sign-in.
Acceptance Criteria	1. Google and GitHub sign-in options are presented. 2. Successful authentication creates a valid session. 3. Protected routes reject unauthenticated access. 4. Sign-out clears the application session.
Priority	Must Have

FR-110: Profile Completion and Organization Access

Field	Detail
Requirement ID	FR-110
Description	New users shall complete profile and organization-access information before entering the shared DAM workspace.
Business Rules	A user may join an existing organization or request creation of a new organization. Organization creation may include storage-mode selection and BYOS configuration. Access shall remain pending until the applicable approval is complete.
Acceptance Criteria	1. The user can supply required profile details. 2. Existing organizations, departments, and roles are selectable where permitted. 3. New-organization requests enter a pending state. 4. Pending, rejected, and suspended states prevent normal workspace access and show a clear explanation.
Priority	Must Have

FR-120: Organization Workspace Provisioning

Field	Detail
Requirement ID	FR-120
Description	The system shall provide each approved organization with an isolated logical workspace and shared asset bucket.
Business Rules	Asset, user, department, taxonomy, sharing, permission, and storage records shall remain scoped to the active organization. The workspace shall expose only authorized organization content.
Acceptance Criteria	1. An approved organization can enter its DAM workspace. 2. Users cannot access records from another organization. 3. The organization bucket is created or resolved consistently.
Priority	Must Have

B. DAM Portal - Asset Library & Ingestion**FR-200: Asset Library Browsing**

Field	Detail
Requirement ID	FR-200
Description	Users shall browse the organization asset library through folders and configurable visual layouts.
Business Rules	The library shall support grid, compact-grid, list, and table views. Directory state, breadcrumbs, sorting, selection, loading, empty, and error states shall be clear.
Acceptance Criteria	1. Users can navigate the directory tree and breadcrumbs. 2. Users can switch views without losing the current folder. 3. Assets can be sorted by name, creation date, or modification date. 4. Selection supports permitted bulk actions.
Priority	Must Have

FR-210: File and Batch Upload

Field	Detail
Requirement ID	FR-210
Description	Authorized users shall upload one or more files to a selected workspace destination.
Business Rules	Upload permission, destination authorization, file policy, filename policy, and storage routing shall be validated before persistence. Mixed batch outcomes shall identify the affected file.
Acceptance Criteria	1. The user can select or drag files. 2. A destination is explicitly selected when required. 3. Progress and completion feedback are shown. 4. Rejected files display an actionable reason. 5. Successful assets appear in the chosen folder.
Priority	Must Have

FR-220: Directory-Tree Upload

Field	Detail
Requirement ID	FR-220
Description	Authorized users shall upload a local directory while preserving its nested folder structure, including empty folders.
Business Rules	Relative paths shall be normalized, unsafe traversal segments rejected, folder nomenclature preflighted, and folders created before dependent file uploads.
Acceptance Criteria	1. The selected hierarchy is recreated beneath the chosen destination. 2. Empty directories are retained. 3. Invalid folder names stop the upload before partial hierarchy creation where practical. 4. Deep nesting remains navigable in the directory tree.
Priority	Must Have

FR-230: Department-Targeted Upload Routing

Field	Detail
Requirement ID	FR-230
Description	Upload destinations shall respect the user's accessible departments and authorized workspace folders.
Business Rules	Users shall not be able to upload into a department or folder outside their calculated access scope. Google Drive destinations shall use stable provider folder identifiers rather than display names alone.
Acceptance Criteria	1. Only authorized destinations are listed. 2. Nested folder selection is supported. 3. Unauthorized destination identifiers are rejected server-side. 4. Uploaded assets are written to the selected physical and logical location.
Priority	Must Have

FR-240: File Nomenclature and Allowed Formats

Field	Detail
Requirement ID	FR-240
Description	The system shall enforce department-specific file naming templates and allowed file extensions when the organization enables nomenclature governance.
Business Rules	Templates shall be assembled from ordered segments. Segments may define controlled values. Segment keys and values shall not introduce the underscore delimiter. File extensions shall be normalized and validated independently of the filename stem.

Field	Detail
Acceptance Criteria	1. Administrators can define ordered naming segments and allowed extensions. 2. Uploaders receive a guided nomenclature form. 3. Non-compliant names or extensions are rejected with a useful message. 4. Compliant final names are used consistently across storage and metadata.
Priority	Must Have

FR-250: Folder Nomenclature

Field	Detail
Requirement ID	FR-250
Description	The system shall support department-specific folder naming templates for folder creation and directory upload.
Business Rules	Folder policies shall use ordered segments and optional controlled values. Directory uploads shall validate every relative folder component that is subject to the policy.
Acceptance Criteria	1. Administrators can define folder segments separately from file segments. 2. Folder requests and direct creation use the effective policy. 3. Directory preflight returns the precise invalid path and expected format.
Priority	Must Have

FR-260: Duplicate and Filename Collision Handling

Field	Detail
Requirement ID	FR-260
Description	The system shall detect duplicate content and avoid unintended overwrites when a target filename already exists.
Business Rules	Content hashes shall be used where available to identify matching content. Name collisions shall produce a deterministic alternative name or a defined duplicate response. Existing assets shall not be silently overwritten.
Acceptance Criteria	1. Identical content can be linked to its original asset record. 2. A colliding filename is resolved predictably. 3. The user receives the final stored name. 4. Storage and database references remain consistent.
Priority	Must Have

FR-270: Folder Creation and Requests

Field	Detail
Requirement ID	FR-270

Field	Detail
Description	Folder creation shall follow the organization's hierarchy policy and the user's permission level.
Business Rules	Users with direct folder-creation authority may create compliant folders. Other users shall submit a request containing the proposed name and parent location. Authorized reviewers may approve or reject with a note; approval shall create the final folder.
Acceptance Criteria	1. Direct creation is limited to permitted users. 2. Other users can submit a request. 3. Reviewers see pending and historical requests. 4. Approval creates the folder once. 5. Rejection records the reviewer outcome.
Priority	Must Have

FR-280: Template Folder Structures

Field	Detail
Requirement ID	FR-280
Description	Organization administrators shall create predefined business folder structures from approved templates.
Business Rules	Templates may include Clients, Onboarding, Projects, Finance, and HR structures. Client onboarding may create Brand Assets, Corporate, Knowledge Transfer, Monthly Reports, Projects, and Social Media Calendar subfolders.
Acceptance Criteria	1. Available templates are shown with descriptions. 2. Selecting a template creates the complete structure once. 3. Duplicate root structures are prevented or safely reused. 4. Created folders appear in the organization library.
Priority	Should Have

C. DAM Portal - Asset Actions, Preview & Collections

FR-300: Core Asset Actions

Field	Detail
Requirement ID	FR-300
Description	Authorized users shall open, download, rename, copy, move, and move assets to Trash.
Business Rules	Each operation shall validate tenant, department, object, destination, and action permissions. Folder moves shall prevent cycles and invalid descendants. Destructive actions shall use soft deletion where defined.
Acceptance Criteria	1. Only permitted actions are displayed and accepted. 2. Rename and move update logical and physical references consistently. 3. Copy creates a distinct managed asset. 4. Trash removes the item from normal views without immediate hard deletion.
Priority	Must Have

FR-310: File Preview

Field	Detail
Requirement ID	FR-310
Description	Users shall preview supported assets without downloading them first.
Business Rules	Preview shall support common images, PDF, DOCX, XLSX, PPTX, video, audio, and text-compatible content through an appropriate renderer. Remote content shall be delivered through authorized server routes when direct provider URLs are unsuitable.
Acceptance Criteria	1. Supported files open in an in-app preview. 2. Preview is available from the asset page and floating preview surface. 3. Unsupported types show a graceful fallback and download option. 4. Preview failure does not corrupt the library state.
Priority	Must Have

FR-320: Asset Information and Metadata Editing

Field	Detail
Requirement ID	FR-320
Description	Users shall view technical asset information and authorized users shall edit tags and custom metadata.
Business Rules	Core properties include name, type, content type, size, dimensions where available, path, storage reference, timestamps, visibility, and ownership context. Custom fields shall follow the organization's taxonomy definition.
Acceptance Criteria	1. Asset information is displayed in a consistent details panel. 2. Tags and custom fields can be saved by permitted users. 3. Required and controlled taxonomy fields are validated. 4. Updated metadata is available to filters and search.
Priority	Must Have

FR-330: Favorites

Field	Detail
Requirement ID	FR-330
Description	Users shall add or remove assets from a personal Favorites collection.
Business Rules	Favorite membership shall be unique per user and asset and remain organization-scoped.
Acceptance Criteria	1. Users can favorite one or multiple assets. 2. The Favorites view displays the user's current collection. 3. Removing a favorite does not delete the asset.
Priority	Should Have

FR-340: Recent, Shared, Published and Trash Views

Field	Detail
Requirement ID	FR-340
Description	The DAM Portal shall provide dedicated operational views for recent activity, assets shared with the user, published assets, and deleted assets.
Business Rules	Each view shall apply its own query criteria while preserving tenant and permission filters. Trash shall support restoration to an appropriate location.
Acceptance Criteria	1. Recent assets are ordered by latest activity. 2. Shared shows assets explicitly granted to the user. 3. Published shows assets with publication records. 4. Trash shows soft-deleted items and supports permitted restoration.
Priority	Must Have

FR-350: Asset Sharing

Field	Detail
Requirement ID	FR-350
Description	Authorized users shall share assets or folders with organization users using viewer or editor access.
Business Rules	Shares shall be unique per recipient and asset. Sharing shall not bypass organization boundaries or higher-level restrictions. Share counts shall remain synchronized with share records.
Acceptance Criteria	1. The sharing surface lists current recipients. 2. Users can be added with viewer or editor access. 3. Recipients see the item in Shared. 4. Removing access removes it from the recipient's Shared view.
Priority	Must Have

FR-360: Asset Publishing

Field	Detail
Requirement ID	FR-360
Description	Authorized users shall publish selected assets or folders through a managed public route and domain mapping.
Business Rules	Publication shall be explicit, unique for the published object, and reversible. Public access shall expose only the published content and required descendants.
Acceptance Criteria	1. A permitted user can publish an asset or folder. 2. The system creates a stable public reference. 3. Published content appears in the Published view. 4. Unpublishing removes public availability.
Priority	Should Have

D. DAM Portal - Search & RAG

FR-400: Keyword Search

Field	Detail
Requirement ID	FR-400
Description	Users shall search authorized assets by filename and available indexed metadata.
Business Rules	Results shall be restricted to the user's organization, accessible departments, and asset permissions. Search shall support direct navigation to the result location.
Acceptance Criteria	1. Entering a query returns matching authorized assets. 2. Results show name, path, type, size, and modified date where available. 3. Selecting a result opens or locates the asset.
Priority	Must Have

FR-410: Filtering and Faceted Search

Field	Detail
Requirement ID	FR-410
Description	Users shall narrow asset results by available properties, tags, custom taxonomy values, and sharing state.
Business Rules	Filter values shall be derived from organization-authorized content and configured taxonomies. Multiple compatible filters may be combined.
Acceptance Criteria	1. Filters are available for relevant type, metadata, and sharing fields. 2. Applying filters updates the visible result set. 3. Clearing filters restores the unfiltered view.
Priority	Must Have

FR-420: Smart and Semantic Search

Field	Detail
Requirement ID	FR-420
Description	The search experience shall combine keyword matches with semantic results when RAG index data is available.
Business Rules	Results shall identify semantic and keyword match types. Semantic retrieval shall never return an asset that the requesting user cannot access. The interface shall degrade to keyword search when semantic services are unavailable.
Acceptance Criteria	1. Search can present combined and separated result groups. 2. Match type is visible. 3. Semantic service failure leaves keyword search usable. 4. Authorization is applied before results are returned.
Priority	Should Have

FR-430: RAG Processing

Field	Detail
Requirement ID	FR-430
Description	Authorized users shall submit supported assets for parsing, chunking, embedding, and indexing for retrieval workflows.
Business Rules	Processing shall support configured document, image, presentation, spreadsheet, PDF, and video parsers. A processing artifact and job state shall be associated with the source asset. Failures shall identify the stage without exposing secrets.
Acceptance Criteria	1. The RAG action is limited to users with <code>canUseRag</code> . 2. Processing progress is displayed in a modal or equivalent surface. 3. Completion indicates the created artifact or index outcome. 4. Failure provides a recoverable message and does not alter the source file.
Priority	Must Have

E. DAM Portal - Organization Governance**FR-500: Department Hierarchy**

Field	Detail
Requirement ID	FR-500
Description	Organization administrators shall define departments and optional nested sub-teams, with each department mapped to a logical or provider folder.
Business Rules	Department names shall be unique within the relevant organization context. Parent-child relationships shall not form cycles. Google Drive organizations may map departments to provider folder identifiers.
Acceptance Criteria	1. Admins can create and view departments and sub-teams. 2. Department records are organization-scoped. 3. Folder mappings are retained. 4. The hierarchy is available to access control, uploads, invitations, and nomenclature.
Priority	Must Have

FR-510: Role-Based Access Control

Field	Detail
Requirement ID	FR-510
Description	The DAM Portal shall enforce hierarchical roles for Admin, Department Head, Team Lead, Team Member, and Intern.
Business Rules	Admin has organization-wide authority. Department Heads manage permitted users and rules within their own department and cannot assign peer-or-higher roles. Lower roles receive only configured permissions.

Field	Detail
Acceptance Criteria	1. Every active user has a valid role. 2. Management actions respect role rank and organization membership. 3. Department Heads cannot manage users outside their department or elevate users to Department Head/Admin. 4. Unauthorized API calls are rejected.
Priority	Must Have

FR-520: Permission Resolution and Overrides

Field	Detail
Requirement ID	FR-520
Description	File-service permissions shall be resolved from Admin authority, individual overrides, department role defaults, and organization-wide role defaults.
Business Rules	The permission order shall be: Admin allow; explicit individual decision; department-role decision; global-role decision; otherwise deny. Permissions include view, download, upload, create folder, rename, delete, share, edit metadata, and use RAG.
Acceptance Criteria	1. Admin receives required organization access. 2. Explicit user overrides take precedence. 3. Department defaults take precedence over global defaults. 4. Missing permission data fails closed. 5. UI visibility and API enforcement remain aligned.
Priority	Must Have

FR-530: Department Access Scope

Field	Detail
Requirement ID	FR-530
Description	Users shall access their own department plus explicitly granted departments, or all departments when authorized.
Business Rules	Admin and all-department access provide the complete organization department set. Inter-department visibility rules may further govern what ordinary users can browse.
Acceptance Criteria	1. Accessible departments are calculated consistently. 2. Grants do not cross organization boundaries. 3. Folder listing, upload destinations, search, and file operations use the same scope.
Priority	Must Have

FR-540: User Administration and Approval

Field	Detail
Requirement ID	FR-540

Field	Detail
Description	Authorized organization users shall review pending members and manage eligible users within their scope.
Business Rules	Users may have pending, active, rejected, or suspended access states as applicable. Role, department, and approval changes shall respect hierarchy rules. Department Heads shall manage only lower-ranked users in their own department.
Acceptance Criteria	1. Pending users are visible to authorized reviewers. 2. Approval assigns valid organization, role, and department context. 3. Rejected or suspended users cannot enter the workspace. 4. User edits are auditable.
Priority	Must Have

FR-550: Department Invitations

Field	Detail
Requirement ID	FR-550
Description	Organization administrators shall generate time-limited invitations for Department Heads.
Business Rules	An invitation shall bind organization, department, email, token, status, and expiry. A token shall be single-use after successful acceptance.
Acceptance Criteria	1. Admin can select a department and invitee email. 2. The system generates a unique invitation link. 3. Expired or accepted tokens cannot be reused. 4. Acceptance assigns the intended organization and department role.
Priority	Should Have

FR-560: Taxonomy Configuration

Field	Detail
Requirement ID	FR-560
Description	Administrators shall define organization-wide or department-specific custom metadata fields.
Business Rules	A field has a unique key, name, type (text , select , or multiselect), optional controlled options, required flag, and optional department scope.
Acceptance Criteria	1. Admin can create, edit, and delete taxonomy fields. 2. Duplicate keys are rejected within the organization. 3. Asset metadata editing renders the correct control type. 4. Required fields and allowed options are validated.
Priority	Must Have

FR-570: Organization Settings and Governance Rules

Field	Detail
-------	--------

Field	Detail
Requirement ID	FR-570
Description	Organization administrators shall configure departments, custom role names where supported, and governance rules.
Business Rules	Governance controls include nomenclature enforcement, folder-hierarchy enforcement, and inter-department visibility. Default role identities shall remain protected from invalid deletion or duplication.
Acceptance Criteria	1. Settings load only for the active organization. 2. Changes persist and are reflected in the session where necessary. 3. Invalid role or rule values are rejected. 4. Governance changes affect subsequent relevant operations.
Priority	Must Have

F. DAM Portal - Storage Integrations

FR-600: Storage Routing

Field	Detail
Requirement ID	FR-600
Description	Asset operations shall route to the storage backend configured for the organization while maintaining a consistent DAM experience.
Business Rules	Supported modes include platform/local storage, Google Drive, and BYOS object storage. Database asset records shall retain enough information to resolve the physical object without exposing storage credentials.
Acceptance Criteria	1. Upload, preview, download, rename, move, copy, and delete use the organization storage target. 2. Logical paths remain consistent. 3. A backend failure produces an actionable error. 4. No operation writes to another organization's storage.
Priority	Must Have

FR-610: Google Drive Connection and Approval

Field	Detail
Requirement ID	FR-610
Description	Organizations shall connect and select a Google Drive folder, subject to the configured approval workflow.
Business Rules	Access and refresh tokens, token expiry, selected folder identity, request state, user, and organization shall be retained securely. A connection shall not become active before approval where approval is required.
Acceptance Criteria	1. The user authorizes Google access. 2. The user can select an accessible Drive folder. 3. Pending requests show an awaiting-approval state. 4. Approved connections become available to the organization. 5. Expired access tokens are refreshed when a valid refresh token exists.

Field	Detail
Priority	Must Have

FR-620: Google Drive Asset Operations

Field	Detail
Requirement ID	FR-620
Description	The DAM Portal shall list and manage authorized Google Drive files and folders through the same core library experience.
Business Rules	Operations include listing folders, creating folders, uploading, downloading, renaming, deleting, RAG submission, and importing supported links. Provider identifiers shall be used for mutations.
Acceptance Criteria	1. The directory tree lists authorized Drive folders. 2. Uploads arrive in the selected Drive folder. 3. Rename, delete, download, and preview operate on the intended Drive object. 4. Errors from Google are translated into safe user-facing messages.
Priority	Must Have

FR-630: Managed Link Import

Field	Detail
Requirement ID	FR-630
Description	Authorized users shall import a supported remote file link as an actual managed DAM asset.
Business Rules	The server shall retrieve the file, validate the response and size, apply upload and naming policies, and store the content in the target backend. A shortcut-only record shall not satisfy this requirement.
Acceptance Criteria	1. The user supplies a supported link and destination. 2. The actual file is downloaded and stored. 3. The final managed filename is shown. 4. Invalid, inaccessible, or oversized responses are rejected safely.
Priority	Should Have

FR-640: Bring Your Own Storage (BYOS)

Field	Detail
Requirement ID	FR-640
Description	An organization creator shall configure a supported organization-owned object-storage bucket during onboarding.
Business Rules	Supported providers are AWS S3, Cloudflare R2, and Google Cloud Storage using supported service-account or OAuth connection modes. Credentials shall be verified before use and linked to the approved organization.

Field	Detail
Acceptance Criteria	1. Provider-specific fields are presented. 2. Credentials and bucket access are verified. 3. Invalid credentials do not create an active storage configuration. 4. Approved configuration routes organization assets to the selected bucket.
Priority	Must Have

G. Superadmin Portal

FR-700: Superadmin Authentication and Isolation

Field	Detail
Requirement ID	FR-700
Description	The Superadmin Portal shall use a separate protected authentication and session boundary from ordinary DAM users.
Business Rules	Superadmin pages and APIs shall require a valid Superadmin session. Ordinary OAuth sessions shall not grant Superadmin access. Logout shall invalidate the privileged session.
Acceptance Criteria	1. Superadmin has a dedicated login. 2. Invalid credentials are rejected. 3. Direct navigation without a privileged session is blocked. 4. Logout returns the user to Superadmin login.
Priority	Must Have

FR-710: Platform Dashboard

Field	Detail
Requirement ID	FR-710
Description	Superadmin shall view a platform dashboard summarizing organizations, total users, storage used, and active Google Drive connections.
Business Rules	Totals shall be derived from current platform records and shall not expose organization credentials or file content.
Acceptance Criteria	1. Dashboard totals load after privileged authentication. 2. Empty-platform totals display as zero. 3. Failure states do not expose internal stack traces.
Priority	Must Have

FR-720: Organization Request Review

Field	Detail
Requirement ID	FR-720
Description	Superadmin shall review pending organization creation requests and approve or reject them with an optional review note.

Field	Detail
Business Rules	Approval shall create the organization, connect the requesting user, apply the requested storage type, and bind verified BYOS configuration when applicable. A request shall be finalized only once.
Acceptance Criteria	1. Pending requests are listed with requester and requested organization details. 2. Approval provisions the organization and user access. 3. Rejection records the outcome. 4. Repeated finalization is prevented.
Priority	Must Have

FR-730: Organization Lifecycle Administration

Field	Detail
Requirement ID	FR-730
Description	Superadmin shall list organizations, inspect organization detail, suspend or reactivate an organization, and delete an organization when authorized by platform policy.
Business Rules	Suspension shall prevent organization users from normal DAM access without deleting tenant data. Deletion shall require explicit confirmation and follow the approved retention policy.
Acceptance Criteria	1. Organizations can be searched or opened from the list. 2. Suspension immediately changes access state. 3. Reactivation restores eligible access. 4. Deletion targets only the selected organization and returns a clear outcome.
Priority	Must Have

FR-740: Organization Feature Access

Field	Detail
Requirement ID	FR-740
Description	Superadmin shall control whether an organization can use nomenclature rules, department hierarchy, user-permission customization, and template folders.
Business Rules	Feature values shall be stored per organization and used by the DAM Portal to control access to the relevant functions. Disabling a feature shall not silently delete its existing configuration.
Acceptance Criteria	1. Superadmin can view and change each feature control. 2. Changes persist per organization. 3. The DAM Portal respects the current feature set. 4. Existing configuration remains recoverable when re-enabled.
Priority	Must Have

FR-750: Cross-Organization User Administration

Field	Detail
-------	--------

Field	Detail
Requirement ID	FR-750
Description	Superadmin shall list platform users and edit or remove a selected user.
Business Rules	User changes shall identify the target user and organization context. Destructive removal shall use explicit confirmation and shall not affect unrelated users.
Acceptance Criteria	1. The user list includes identity, organization, role, and access state where available. 2. Superadmin can edit supported fields. 3. Superadmin can remove the selected user. 4. All operations require a privileged session.
Priority	Must Have

FR-760: Google Drive Request Administration

Field	Detail
Requirement ID	FR-760
Description	Superadmin shall review organization Google Drive hosting requests and approve or reject them.
Business Rules	Approval shall activate only the selected request and organization connection. Rejection shall preserve the outcome for operational traceability.
Acceptance Criteria	1. Pending Drive requests are listed with organization and folder context. 2. Approval activates the request. 3. Rejection prevents connection use. 4. Repeated processing is prevented.
Priority	Must Have

FR-770: Nomenclature Oversight

Field	Detail
Requirement ID	FR-770
Description	Superadmin shall inspect and update an organization's department-specific DAM nomenclature policy.
Business Rules	Superadmin access shall be explicit to the selected organization and department. The same validation rules used by organization administration shall apply.
Acceptance Criteria	1. Organization departments are selectable. 2. Existing file and folder nomenclature values load correctly. 3. Valid changes persist to the selected organization only. 4. Invalid templates, segments, values, or extensions are rejected.
Priority	Must Have

H. Cross-Portal Requirements

FR-800: Tenant Isolation

Field	Detail
Requirement ID	FR-800
Description	All DAM data and storage operations shall enforce organization isolation, except for explicitly privileged Superadmin operations.
Business Rules	Organization identity shall be derived from the authenticated session or the privileged, validated Superadmin target. Client-supplied organization identifiers alone shall not authorize access.
Acceptance Criteria	1. Cross-tenant asset and configuration requests are denied. 2. Queries include organization scope. 3. Storage paths and credentials are resolved for the authorized organization. 4. Tests cover cross-tenant denial.
Priority	Must Have

FR-810: Permission Change Audit Trail

Field	Detail
Requirement ID	FR-810
Description	Changes to organization role permissions and individual permission overrides shall create an audit record.
Business Rules	Audit data shall identify organization, actor, target user or role, department where applicable, changes, and timestamp. Audit records shall not contain credentials.
Acceptance Criteria	1. A successful permission change creates one audit entry. 2. The actor and target are identifiable. 3. Failed changes do not create misleading success entries. 4. Audit records remain organization-scoped.
Priority	Must Have

FR-820: Consistent Error Handling

Field	Detail
Requirement ID	FR-820
Description	Both portals shall present safe, actionable errors and preserve recoverable user state.
Business Rules	Errors shall distinguish authentication, authorization, validation, missing resource, provider failure, conflict, and unexpected failure. Secrets, tokens, credentials, local paths, and stack traces shall not be exposed to users.
Acceptance Criteria	1. Validation errors identify the field or asset. 2. Permission errors do not reveal hidden resources. 3. Provider failures offer a retry or corrective action. 4. Unexpected failures are logged with a request correlation identifier where configured.

Field	Detail
Priority	Must Have

04 - Non-Functional Requirements

NFR-100: Performance

Measure	Requirement
Standard page interaction	Primary navigation and library shell should become usable within 3 seconds under normal operating conditions.
Folder listing	First page of a typical folder should return within 2 seconds, excluding provider outages.
Search	Keyword results should return within 3 seconds for the expected organization dataset.
Upload feedback	Progress feedback shall begin within 1 second of upload initiation.
Large libraries	Directory and file APIs shall use pagination or bounded result sets to avoid loading an entire organization tree at once.
RAG processing	Long-running parsing and embedding shall execute asynchronously and expose progress rather than block the request lifecycle.

NFR-200: Availability and Reliability

- Monthly availability target: **[Stakeholder input required]**.
- Database and asset-storage health shall be independently observable.
- Upload workflows shall not create a successful database record when the physical asset was not stored.
- Storage mutations and metadata updates shall use compensating cleanup or reconciliation when one side fails.
- OAuth token refresh shall be attempted before declaring an eligible Google Drive session invalid.
- Production data shall have documented backup and restore procedures appropriate to the selected runtime.
- Critical startup configuration shall be validated before the service accepts traffic.

NFR-300: Security

- All protected APIs shall require an authenticated user or Superadmin session.
- Authorization shall be enforced server-side for every protected operation.
- Tenant and department scope shall be applied before resource return or mutation.
- Superadmin credentials and sessions shall remain separate from ordinary user OAuth sessions.
- OAuth tokens, BYOS keys, service-account private keys, and session secrets shall never be returned to the browser after storage.
- Production traffic shall use TLS.
- File paths and remote URLs shall be normalized and validated to prevent traversal and server-side request misuse.
- Uploads shall enforce configured type and size limits.
- Public routes shall expose only explicitly published resources.

- Logs shall redact secrets and sensitive authorization headers.

NFR-400: Usability and Accessibility

- Both portals shall provide clear loading, empty, success, warning, and failure states.
- Core DAM functions shall be usable on desktop and mobile layouts.
- Keyboard users shall be able to reach navigation, menus, dialogs, asset actions, and forms.
- Interactive controls shall have discernible names and visible focus states.
- Text and controls shall maintain adequate contrast in light and dark modes.
- Forms shall associate labels, descriptions, validation messages, and required states with their fields.
- Deep folder nesting shall remain readable without losing the selected context.

NFR-500: Scalability

- Organization queries shall use indexed tenant, parent, name, status, and user relationships.
- Folder and search APIs shall support pagination.
- File content shall remain in object/file storage rather than the relational database.
- RAG jobs shall be separable from interactive web requests.
- Storage backends shall be selected per organization without requiring a different user experience.
- Deployment shall support movement from a small persistent VM to a larger node or managed storage configuration without changing business records.

NFR-600: Maintainability and Testability

- Shared policy functions shall define permissions, upload routing, nomenclature, collision handling, pagination, and storage selection.
- Client and server code shall use stable API contracts for file and folder identifiers.
- Automated contract tests shall cover access control, upload governance, directory trees, storage routing, preview, RAG artifacts, OAuth refresh, nomenclature, and Superadmin operations.
- Deployment configuration shall fail early when mandatory environment values are missing.
- Schema migrations shall be repeatable and version-controlled.

NFR-700: Data Protection and Retention

- Tenant data shall be retained according to the organization's approved retention policy: **[Stakeholder input required]**.
- Soft-deleted assets shall remain recoverable for **[Stakeholder input required]** days before physical deletion.
- Organization deletion shall follow an approved workflow covering files, provider objects, metadata, users, credentials, and backups.
- OAuth and BYOS credentials shall be removed or invalidated when the associated organization is permanently deleted.
- Backup retention and recovery-point objectives shall be defined by deployment environment.

05 - Current State Analysis

As-Is Operating Model

The application is a Nuxt 3 / Vue 3 DAM with Nitro server APIs, Drizzle ORM, and an organization-scoped data model. The user-facing portal combines the asset library with organization administration and department-governance pages. A separate layout, middleware boundary, login, and API namespace support Superadmin operations.

The asset library uses a shared organization bucket, directory-tree navigation, asset collections, file action menus, preview components, search, upload orchestration, and storage-specific APIs. Organization governance is represented through roles, organization/department permissions, individual overrides, department grants, naming policies, taxonomy definitions, folder requests, invitation records, and organization settings.

System Context

Layer	Current Product Responsibility
Web client	Vue/Nuxt portal views, forms, previews, navigation, and progress feedback
Nitro API	Authentication, authorization, validation, business rules, provider routing, and response shaping
Relational data	Organizations, departments, users, permissions, assets, metadata, sharing, governance, and storage configuration
Asset storage	Platform/local file storage, Google Drive, or organization BYOS object storage
RAG services	Content parsing, progress tracking, embedding, and vector index integration
External identity	Google and GitHub OAuth
Platform operations	Separate Superadmin session, pages, and APIs

Existing Integrations

- Google OAuth and GitHub OAuth for DAM users.
- Google Drive OAuth, folder selection, listing, and asset operations.
- AWS S3-compatible storage for AWS S3 and Cloudflare R2.
- Google Cloud Storage through service-account or OAuth-supported modes.
- Local/persistent filesystem storage for Node/GCP runtime deployments.
- NuxtHub / Cloudflare-compatible blob and database services where configured.
- Pinecone-compatible vector indexing and external parsing/model services for RAG.

Business and Operational Constraints

1. Storage behavior must remain consistent despite different provider identifiers and URL behaviors.
2. Organization, department, role, and user-specific access rules can intersect on a single operation.

3. Directory uploads must validate hierarchy before creating provider folders.
4. Google OAuth tokens may expire during normal operation and require refresh handling.
5. Document preview and RAG parsing depend on file type, file availability, and runtime resources.
6. Small deployment targets require bounded memory usage, especially during builds, uploads, previews, and document processing.

06 - Future-State Operating Design

Target Operating Principles

1. **One governed workspace per organization** - the DAM is the authoritative organizational asset surface.
2. **Policy before persistence** - permissions, destination, naming, format, path, and collision rules are evaluated before content is committed.
3. **Provider-neutral experience** - users perform consistent asset tasks regardless of the configured storage backend.
4. **Server-authoritative security** - client controls improve usability, while APIs remain the final authorization boundary.
5. **Department-aware by default** - asset navigation and operations use the user's calculated department scope.
6. **Operational separation** - platform-level Superadmin responsibilities stay outside organization administration.
7. **Recoverable long-running work** - uploads and RAG processing provide progress, durable state where required, and clear failure recovery.

Target DAM Portal Flow

1. User authenticates through Google or GitHub.
2. The system resolves profile, organization, role, department, approval, and organization status.
3. The user enters the organization workspace and sees only authorized folders and collections.
4. Upload or file actions are evaluated through permission, department, governance, and storage policies.
5. Asset metadata is persisted in the DAM database and content is written to the organization's configured storage.
6. Authorized users search, preview, download, share, publish, or process the asset for RAG.
7. Administrators and Department Heads configure governance within their defined scope.

Target Superadmin Flow

1. Superadmin authenticates through the privileged login.
2. Platform dashboard and request queues load from privileged APIs.
3. Superadmin approves or rejects organization and Google Drive requests.
4. Superadmin administers the selected organization, users, feature access, and nomenclature policy.
5. Organization suspension or reactivation immediately affects DAM access.
6. Privileged operations are logged by the platform's operational logging controls.

Change Management Approach

- Confirm document owner, business sponsor, technical owner, and security approver.
- Convert each requirement into testable release criteria.
- Validate storage modes independently and through the common asset-action contract.

- Pilot governance rules with a representative organization and department hierarchy.
- Provide organization-admin guidance for permissions, nomenclature, taxonomy, and folder templates.
- Provide operations guidance for request approval, tenant suspension, backup, and restore.
- Review success measures after the agreed stabilization period.

07 - Workflow Diagrams & User Journeys

Journey 1: Organization Onboarding

OAuth sign-in -> Profile completion -> Join existing organization OR request new organization -> Select storage configuration -> Superadmin review -> Organization provisioned -> User session refreshed -> DAM workspace access

Business Rules

- New organizations require Superadmin approval.
- BYOS configuration must be verified before it becomes the organization's active storage target.
- Rejected, pending, or suspended access states cannot enter the library.

Journey 2: Governed Asset Upload

Choose files/folder -> Select authorized destination -> Resolve department policy -> Validate extensions -> Build/validate nomenclature -> Preflight directory paths -> Detect duplicate/collision -> Store content -> Persist metadata -> Refresh library -> Optional RAG processing

Alternate Outcomes

- No upload permission: stop before file transfer.
- Invalid destination: refresh available destination list.
- Naming violation: show required template and invalid segment.
- Duplicate content: link or report the existing asset according to collision policy.
- Provider failure: preserve retrievable state and do not report completion.

Journey 3: Folder Request

User proposes folder -> System applies folder nomenclature -> Request stored as pending -> Admin/Department Head reviews -> Approve and create folder OR reject with note -> Request history updated

Journey 4: Search and Retrieval

Enter query -> Apply organization/department scope -> Run keyword search -> Add semantic results when available -> Apply facets -> Open preview -> Download or perform permitted action

Journey 5: Google Drive Organization

Authorize Google -> Select Drive folder -> Request pending -> Superadmin approves -> Map organization/departments -> Browse Drive hierarchy -> Upload/manage through DAM -> Refresh token when necessary

Journey 6: Superadmin Organization Administration

Privileged login -> Platform dashboard -> Select request or organization -> Review data -> Approve/reject or administer -> Persist selected tenant change -> Return clear outcome

Cross-Functional Responsibility Matrix

Activity	Member	Team Lead	Department Head	Org Admin	Superadmin	System
Authenticate and complete profile	R	R	R	R	-	A
Upload and manage permitted assets	R	R	R	R	-	A
Request governed folders	R	R	R	R	-	A
Review department folder requests	-	-	R	R	-	A
Configure department permissions	-	-	R	R	-	A
Configure organization rules	-	-	C	R	C	A
Approve organization creation	-	-	-	-	R	A
Approve Google Drive hosting	-	-	-	C	R	A

Activity	Member	Team Lead	Department Head	Org Admin	Superadmin	System
Control organization feature access	-	-	-	C	R	A

Legend: **R** = responsible, **A** = automated/system execution, **C** = consulted or affected.

State Transitions

User Access

Pending -> Active -> Suspended -> Active

Pending -> Rejected

Organization

Requested -> Approved/Active -> Suspended -> Active

Requested -> Rejected

Folder Request

Pending -> Approved

Pending -> Rejected

Google Drive Request

Pending -> Approved

Pending -> Rejected

Asset

Created -> Active -> Trashed -> Restored

Created/Active -> Published -> Unpublished

08 - Data Flow & Data Requirements

Level 0 Context

External Entity	Data Sent to DAM	Data Received from DAM
DAM User	Identity response, profile, uploads, searches, metadata, actions	Authorized library, previews, results, progress, outcomes
Organization Admin	Departments, roles, permissions, taxonomy, governance	Organization configuration and user state
Superadmin	Privileged credentials and administration decisions	Platform totals, requests, organizations, users, policies
OAuth Providers	Identity claims and tokens	OAuth authorization request
Google Drive	Folder/file data and tokens	Authorized Drive operations
BYOS Provider	Object responses and provider errors	Verified object-storage requests
RAG Services	Parse/embedding/index results	Authorized source content and processing request

Level 1 Data Flow

1. **Identity and session** - OAuth or Superadmin login creates a protected session.
2. **Access context** - server resolves organization, status, role, department, grants, and permissions.
3. **Library query** - server returns authorized asset metadata and folder structure.
4. **Asset ingestion** - upload policy validates input and routes content to configured storage.
5. **Metadata persistence** - relational records capture asset identity, paths, ownership, metadata, and relationships.
6. **Asset delivery** - preview/download routes read from the provider and return authorized content.
7. **RAG processing** - eligible content is parsed and indexed; progress is returned to the user.
8. **Administration** - DAM administrators update organization-scoped rules; Superadmin updates platform-scoped tenant state.

Core Data Entities

Entity	Business Purpose	Key Relationships
Organization	Tenant boundary and lifecycle state	Departments, users, buckets, features, rules, assets
Organization Request	Pending organization onboarding decision	Requesting user, BYOS configuration, review outcome
User	Identity, role, department, and access state	Organization, department, shares, favorites, permissions
Department	Hierarchical business unit and folder mapping	Organization, parent department, users, nomenclature

Entity	Business Purpose	Key Relationships
Department Invite	Time-limited Department Head onboarding	Organization, department, email, token
Bucket	Logical organization asset container	Organization, files
File	Asset or folder metadata and storage reference	Bucket, parent folder, user, organization, shares, favorites
Favorite	Personal saved-asset relationship	User, file, organization
Share	Viewer/editor access relationship	Recipient user, file, organization
Website Publication	Published route/domain mapping	File or folder, bucket
Taxonomy	Custom metadata field definition	Organization, optional department
Nomenclature	File/folder naming and format policy	Organization, department, updating user
Folder Request	Governed folder-creation decision	Requester, reviewer, department, target parent
Role Permission	Role-based service-action defaults	Organization, department/global scope, role
User Permission Override	Explicit per-user service-action decisions	User, organization, department
Department Access Grant	Additional department visibility	User, organization, department
Permission Audit Log	Trace of permission configuration changes	Actor, target, organization, department
Google Drive Folder	Drive authorization and selected folder state	User, organization, provider folder
BYOS Configuration	Organization object-storage connection	Requesting user, organization, provider, bucket
Google Drive Rules	Organization governance toggles	Organization

Data Quality Rules

- Organization names and user emails shall be unique where enforced by the schema.
- Asset paths shall not collide within the applicable active-file namespace.
- User/asset favorites and user/asset shares shall not be duplicated.
- Taxonomy keys shall be unique within an organization.
- Nomenclature shall have at most one effective record per department.
- Department access grants shall be unique per user and department.
- Provider identifiers shall not be replaced by display names for mutations.
- Timestamps shall be recorded for creation and modification of governed entities.

- JSON metadata shall be validated before persistence and default safely when malformed legacy values are encountered.

Sensitive Data

The following data requires restricted access and redaction from normal responses and logs:

- OAuth access and refresh tokens.
- Superadmin credentials and session material.
- AWS/R2 access keys and secret keys.
- Google Cloud service-account email and private key.
- Google Cloud OAuth tokens.
- Session secrets and provider client secrets.
- Internal storage paths where disclosure would create security risk.

09 - Technology & Architecture

Technology Stack

Layer	Technology / Service
Frontend	Nuxt 3, Vue 3, TypeScript, Nuxt UI, Tailwind CSS, Iconify
Server	Nitro server engine and Nuxt server routes
Authentication	nuxt-auth-utils, Google OAuth, GitHub OAuth, separate Superadmin session
Data Access	Drizzle ORM
Database	SQLite-compatible runtime; NuxtHub/Cloudflare or persistent Node/GCP deployment modes
Platform Storage	NuxtHub Blob or persistent local filesystem, depending on runtime
External Storage	Google Drive, AWS S3, Cloudflare R2, Google Cloud Storage
Preview	Vue Office renderers for DOCX/XLSX/PPTX, PDF preview, browser media renderers
Metadata	EXIF-compatible extraction library and application-defined asset metadata
RAG	Python parsers, embedding workflow, Pinecone-compatible vector store
Deployment	Cloudflare-compatible deployment or Dockerized Node server on GCP/GCE

Conceptual Component Architecture

Component	Responsibility
DAM Web App	Organization library, asset actions, previews, search, uploads, and governance UI
Superadmin Web App	Platform dashboard, request queues, organizations, users, feature controls, nomenclature oversight
Auth Middleware	Session resolution, route protection, organization and access-state gating
Access Policy Layer	Role hierarchy, service permissions, user overrides, department scope
Upload Policy Layer	Destination, file/folder nomenclature, file formats, directory preflight, collision handling
Asset Service	CRUD actions, collections, metadata, sharing, publishing, and provider-neutral responses
Storage Adapters	Platform/local, Google Drive, and BYOS object operations
RAG Pipeline	Parse, chunk, embed, index, progress, and artifact reporting
Administration Services	Organization, user, department, invitation, taxonomy, and request management
Persistence Layer	Drizzle models and runtime database adapter

Deployment Requirements

- Production configuration shall provide strong session secrets and OAuth credentials.
- Persistent Node/GCP deployments shall validate writable database and asset paths at startup.
- Backup and restore scripts shall target explicit configured data directories.
- Small-VM builds shall use controlled memory settings and a production build workflow appropriate to available RAM.
- Container health checks shall test application readiness without requiring a privileged user session.
- Secrets shall be supplied through environment or platform secret management and excluded from source control.

Architecture Decisions

1. The relational database stores asset metadata and governance relationships; binary content remains in storage providers.
2. Storage selection is resolved per organization.
3. User authorization is calculated from role, department, grants, and overrides.
4. The Superadmin Portal uses separate middleware, layout, session, and API routes.
5. Long-running RAG processing uses a job/progress model rather than a single blocking browser request.
6. Provider-neutral asset URLs are served through controlled application routes when direct remote URLs cannot meet authentication or browser requirements.

10 - Project Plan, Governance & Quality

Delivery Workstreams

Workstream	Scope	Exit Criteria
Foundation	Authentication, organization context, database, storage adapters	Approved user can enter an isolated organization workspace
Asset Library	Folder navigation, views, upload, preview, file actions, collections	Core asset lifecycle passes acceptance tests
Governance	Departments, RBAC, overrides, folder requests, nomenclature, taxonomy	Restricted operations follow configured organization policy
Integrations	Google Drive, BYOS, remote import, provider-neutral delivery	Supported storage modes pass the same asset-action contract
RAG	Parsing, progress, artifacts, semantic retrieval	Supported content can be processed and retrieved by authorized users
Superadmin	Privileged login, dashboard, requests, organizations, users, feature controls	Platform operator can manage the tenant lifecycle securely
Production Readiness	Deployment, startup validation, backup/restore, security, performance	Release checklist and recovery rehearsal are approved

Governance Roles

Governance Role	Accountability
Executive Sponsor	Business case, funding, scope and priority decisions
Product Owner	Requirement ownership, acceptance, and backlog decisions
Technical Lead	Architecture, implementation quality, integrations, and technical risk
Security Owner	Identity, authorization, secrets, tenant isolation, and security acceptance
Operations Owner	Deployment, monitoring, backup, restore, and incident readiness
Organization Pilot Owner	Validates real-world folder, permissions, storage, and naming policies

Change Control

1. Proposed change is recorded with business reason, affected requirements, risk, and effort.
2. Product and technical owners assess scope and dependencies.
3. Security reviews changes affecting identity, authorization, storage, public access, or credentials.
4. Approved change updates the BRD version, acceptance criteria, and delivery plan.
5. Rejected or deferred change retains its decision record.

Risk Register

ID	Risk	Impact	Mitigation
R-01	Cross-tenant data exposure	Critical	Central tenant-scope utilities, server enforcement, negative access tests
R-02	Provider behavior differs across storage modes	High	Common storage contract and provider-specific integration tests
R-03	Expired Google tokens interrupt operations	High	Refresh-token handling, explicit reconnect path, safe provider errors
R-04	Upload partially succeeds	High	Preflight validation, ordered persistence, cleanup/reconciliation
R-05	Permission configuration becomes inconsistent	High	Deterministic precedence, fail-closed defaults, audit records
R-06	Deep or large folders degrade responsiveness	Medium	Pagination, bounded tree expansion, indexed parent queries
R-07	RAG processing exhausts runtime resources	High	Asynchronous jobs, bounded file sizes, format-specific parsers, resource controls
R-08	BYOS credentials are disclosed	Critical	Restricted fields, encrypted secret handling where available, response redaction
R-09	Organization deletion removes unintended data	Critical	Exact target resolution, explicit confirmation, retention policy, recoverable backup
R-10	Small deployment cannot build or process large files	Medium	Memory-aware build, streaming/bounded processing, documented scaling trigger

Quality Assurance Plan

Functional Testing

- Authentication, profile completion, approval, and suspension states.
- Organization and department isolation.
- Role hierarchy and permission precedence.
- File, batch, and directory upload.
- Nomenclature, extension, collision, and duplicate rules.
- File actions, preview, collections, sharing, publishing, and restoration.
- Taxonomy metadata and filtered search.

- RAG job progress and artifacts.
- Google Drive and BYOS storage behavior.
- Superadmin request, organization, user, feature, and nomenclature controls.

Non-Functional Testing

- Page and API performance under representative organization size.
- Upload limits and large-directory behavior.
- Accessibility review of navigation, dialogs, forms, tables, and keyboard flow.
- Security tests for authentication bypass, cross-tenant access, path traversal, unsafe remote import, and credential exposure.
- Backup and restore rehearsal.
- Runtime startup and health-check validation.

Release Acceptance

- All Must Have requirements have approved test evidence.
- No unresolved critical or high-severity security defect remains.
- Storage targets in the release scope pass the common asset-action test suite.
- Data migration, backup, restore, and rollback procedures are documented.
- Product, technical, security, and operations owners sign off.

11 - Benefits Realisation

Expected Business Benefits

Quantitative Measures

- Reduced average time required to locate an approved asset.
- Reduced rejected or manually corrected uploads through guided nomenclature.
- Reduced duplicate storage through content-hash detection.
- Reduced administrative effort for department folder and permission governance.
- Increased successful reuse of existing assets through search, Favorites, Shared, and Recent views.
- Increased proportion of supported content available for RAG retrieval.

Targets and baselines require stakeholder measurement during pilot planning.

Qualitative Benefits

- Clearer ownership and accountability for organizational assets.
- Consistent naming and folder practices across departments.
- Greater confidence that users see and perform only authorized actions.
- Easier onboarding of departments and users through governed structures.
- Storage flexibility without fragmenting the DAM experience.
- Better operational control through the separate Superadmin Portal.

Benefits Measurement Framework

Benefit	Baseline	Target	Data Source	Review Frequency
Faster asset retrieval	[Required]	[Required]	Search-to-open timing sample	Monthly
Upload compliance	[Required]	[Required]	Upload validation outcomes	Monthly
Duplicate reduction	[Required]	[Required]	Hash/collision records	Quarterly
Folder approval speed	[Required]	[Required]	Folder request timestamps	Monthly
RAG readiness	[Required]	[Required]	Processing job outcomes	Monthly
Tenant onboarding speed	[Required]	[Required]	Organization request timestamps	Quarterly

12 - Appendices

Appendix A - Glossary

Term	Definition
Asset	A managed file or folder represented in the DAM database and storage backend
BYOS	Bring Your Own Storage; organization-owned AWS S3, Cloudflare R2, or Google Cloud Storage
DAM	Digital Asset Management
Department Scope	Set of departments a user can access through role, membership, grants, or all-department permission
Nomenclature	Ordered rules used to construct and validate file or folder names
OAuth	Delegated authentication/authorization protocol used for provider sign-in and Google access
RAG	Retrieval-Augmented Generation; preparation and retrieval of content for AI-assisted knowledge use
RBAC	Role-Based Access Control
Soft Delete	Marking an asset deleted without immediately removing its content permanently
Superadmin	Privileged platform operator with cross-organization administration capabilities
Taxonomy	Organization-defined custom metadata field and controlled vocabulary
Tenant	An organization and its isolated users, configuration, metadata, and assets

Appendix B - Requirement Source Map

The following project areas were used to derive this BRD:

- `app/pages/[bucket]/` and `app/components/App/` - DAM library, collections, navigation, and workspace behavior.
- `app/components/Upload.vue`, `NomenclatureUploadModal.vue`, `FileMenu.vue`, `FilePreview.vue`, `FileInfo.vue`, `SmartSearch.vue` - ingestion, actions, preview, metadata, and search.
- `app/pages/admin/` - organization settings, users, access control, departments, nomenclature, taxonomy, folder requests, Drive requests, and templates.
- `app/pages/dept-head/` - department dashboard, team leads, and permission management.
- `app/pages/superadmin/` and `app/layouts/superadmin.vue` - Superadmin Portal.
- `server/api/` - business APIs for identity, assets, governance, storage, RAG, and Superadmin operations.
- `server/database/schema.ts` - organizations, users, assets, permissions, governance, sharing, and storage entities.

- [shared/utils/](#) and [server/utils/](#) - access, upload, naming, collision, storage, OAuth, and provider policies.
- [docs/superpowers/specs/](#) and [docs/superpowers/plans/](#) - approved design intent for governance, upload routing, directory trees, GCP hosting, RAG, preview, and nomenclature.
- [tests/](#) - business-contract coverage for access control, uploads, naming, storage, Drive, previews, RAG, and Superadmin behavior.

Appendix C - Open Business Decisions

ID	Decision Required	Owner	Due Date
D-01	Confirm executive sponsor, product owner, document owner, and approvers	[Required]	[Required]
D-02	Approve service availability target and support window	[Required]	[Required]
D-03	Define maximum upload size and supported file-type list per environment	[Required]	[Required]
D-04	Define soft-delete, backup, organization-deletion, and credential-retention periods	[Required]	[Required]
D-05	Approve performance test dataset and expected maximum organization scale	[Required]	[Required]
D-06	Set KPI baselines and target values	[Required]	[Required]
D-07	Confirm production storage modes included in the initial release	[Required]	[Required]

Appendix D - Approval and Sign-Off

Role	Name	Decision	Date	Signature / Reference
Executive Sponsor				
Product Owner				
Technical Lead				
Security Owner				
Operations Owner				

End of Document